Eva Martín López

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Experience

UC Irvine, Balsells Research Fellow – Visiting Researcher
Irvine, CA
Jan 2026 – Present

- Conducting Master's thesis research in Professor Pierre Baldi's group
- Applying deep learning to biomedical imaging, focusing on chemoresistance prediction at the single-cell level in breast cancer research
- Developing and training convolutional neural networks for medical image analysis
- Awarded the Balsells Fellowship, a competitive scholarship for Catalan researchers at UCI

Geprom (Telefónica Tech), Software Developer
Castelldefels, ES
Jan 2025 – Feb 2026

- Developed data spaces for secure data exchange in industrial environments using big data technologies and cloud-based systems
- Contributed to industrial automation and digital transformation projects (Industry 4.0)
- Focused on data infrastructure and industrial digitalization

Institute for Bioengineering of Catalonia (IBEC), Research Assistant in Data Science
Barcelona, ES
Jun 2023 – Jan 2025

- Collaborated on research projects related to data analysis, visualization, and machine learning for biomedical research
- Focused on data preprocessing, statistical modeling, and the application of AI in healthcare
- Co-authored a publication in *Critical Care Medicine* on breathing effort and survival in ARDS patients

Janssen (Johnson & Johnson), Internship – Computational Chemistry Dept.
Toledo, ES
Jan 2023 – Jul 2023

- Developed a Python package for the automation of molecular docking simulations
- Worked with Git and Linux in a pharmaceutical R&D environment

Education

MS **Universitat Politècnica de Catalunya (UPC)**, Master's in Data Science
Barcelona, ES
Sep 2024 – Jun 2026

- Thesis at UC Irvine: Deep learning for biomedical imaging (Balsells Fellowship)
- Focus: big data, statistical modeling, process mining, and machine learning

BS **ESCI – Universitat Pompeu Fabra**, Bachelor's in Bioinformatics
Barcelona, ES
Sep 2020 – Jun 2023

- Santander ESCI-UPF Excellence Scholarship (2020–2021 and 2021–2022)

Publications

Association of Breathing Effort With Survival in Patients With Acute Respiratory Distress Syndrome Jul 2025

Critical Care Medicine

DOI: 10.1097/CCM.0000000000006477

Predictive Model for High Flow Nasal Cannula Failure Based on Metabolomics and Clinical Data — Under review

Journal of Critical Care (submitted 2025)

Rapid Quantification of Short-Chain Fatty Acids by Fourier-Transform Infrared Spectroscopy for Microbiota Quality Assessment — Preprint, 2025

bioRxiv

DOI: 10.1101/2025.11.25.690441

Conferences

Conference of the Catalan Society of Intensive and Critical Medicine (SOCMIC)

- Barcelona, Spain (February 2024). Contributed to the publication and received the **Best Communication Award**

Annual Conference of the Spanish Society of Biomedical Engineering (CASEIB)

- Cartagena, Spain (November 2023). Poster presentation

ISOCS Winter School – Chemical Sensing for Biomedical Applications

- Bormio, Italy (2024). Winter school on transitioning chemical sensing from proof of concept to medical devices

Awards and Scholarships

Balsells Fellowship, UC Irvine (2026)

- Competitive scholarship for Catalan researchers to conduct research at the University of California, Irvine

Best Communication Award, SOCMIC Conference (2024)

- Awarded for the publication contribution at the Conference of the Catalan Society of Intensive and Critical Medicine

Santander ESCI-UPF Excellence Scholarship (2020–2021, 2021–2022)

- Merit-based scholarship for academic excellence

Additional Training

- **Summer School at UPC (2023):** Applied Machine Learning to Solve Real-Life Problems; Web Application Development with Shiny
- **Summer School at Eigenvector Research (2023):** MATLAB Course in Machine Learning
- **Classical Ballet Conservatory:** Institut del Teatre, Barcelona (2014–2017)

Languages

- **Spanish and Catalan:** Native
- **English:** Advanced

Technologies

- **Programming:** Python, R, SQL, MATLAB, Bash
- **Libraries & Frameworks:** PyTorch, Scikit-learn, NumPy, Pandas, Shiny
- **Tools & Platforms:** Git, Linux, Docker, Cloud Technologies, LaTeX
- **Areas:** Machine Learning, Deep Learning (CNNs), Statistical Modeling, Data Preprocessing, Biomedical Imaging